

Jacob Waltermán

jacob.gray.waltermán@gmail.com

Education

- **Bachelor of Arts in Mathematics, Minor in Computer Science**, Western Kentucky University, Bowling Green, KY December 2024
Concentration: Applied Mathematics
GPA: 3.80
Thesis Supervisor: Prof. Ahmet Özkan Özer
Grading Committee: Prof. Molly Dunkum, Prof. Atilla Por
Senior Project: "Convergency and Exponential Stability of an Order-reduced Finite Differences Algorithm for a Boundary-damped Piezoelectric Beam Model"

Awards

- (\$400) **Dr. Wanda Jo Weidemann Scholarship**, 2021-2022, 2023-2024, 2024-2025.
- (\$150) **Johnson, Hugh, and Katherine Mathematics Award**, 2023-2024.
- (\$700) **Honors Diversity Scholarship**, 2021-2022, 2022-2023, 2023-2024.
- (\$4000) **FUSE Grant**, 2022-2024.
- (\$4050) **KEES**, 2020-2024.
- **Governor's Scholar**, <https://gsp.ky.gov>, 2019.

Publications

1. A.K. Aydın, A.Ö. Özer, **J. Waltermán**, *A Novel Finite Difference-Based Model Reduction and a Sensor Design for a Multilayer Smart Beam With Arbitrary Number of Layers*, IEEE Control Systems Letters, vol. 7, pp. 1548-1553, 2023. <https://ieeexplore.ieee.org/document/10121790>
2. A.Ö. Özer, **J. Waltermán**, *Hybrid Power Transmission Line Model: Exact Observability via Sensors at Discontinuous Joints with Uniformly Observable Approximations*, 2025 European Control Conference (ECC), Thessaloniki, Greece, pp. 1648-1653, 2025. <https://doi.org/10.23919/ECC65951.2025.11187295>
3. A.Ö. Özer, **J. Waltermán**, *Exponential Stabilization in a Hybrid Power Transmission Line Model Using Feedback Sensors at Discontinuous Joints and Uniformly Stable Approximations*, Evolution Equations and Control Theory, 2026. <https://doi.org/10.3934/eect.2026007>
4. A.Ö. Özer, A.K. Aydın, **J. Waltermán**, *Robust Model Reductions for the Boundary Feedback Stabilization of Magnetizable Piezoelectric Beams*, Evolution Equations and Control Theory, 2026. <https://doi.org/10.3934/eect.2026041>
5. **J. Waltermán**, A.K. Aydın, S. Leveridge, A.Ö. Özer, *Dynamics of a Longitudinal Piezoelectric Beam*, Wolfram Demonstrations Project, Published: February 7, 2023. <https://demonstrations.wolfram.com/DynamicsOfALongitudinalPiezoelectricBeam/>
6. A.K. Aydın, M. Poynter, A.Ö. Özer, **J. Waltermán**, *Stabilization of the Wave Equation by Direct Fourier Filtering*, Wolfram Demonstrations Project, Published: August 19, 2024. <https://demonstrations.wolfram.com/StabilizationOfTheWaveEquationByDirectFourierFiltering/>
7. A.K. Aydın, A.Ö. Özer, **J. Waltermán**, *Stabilization of a Smart Beam with a Tip Mass*, Wolfram Demonstrations Project, Under Review, 2024.
8. A.K. Aydın, A.Ö. Özer, **J. Waltermán**, *Rapid Stabilization of Heat and Structure Interactions with Boundary Feedback Controllers*, Wolfram Demonstrations Project, Under Review, 2024.

Codes/Projects

1. A.K. Aydın, A.Ö. Özer, Uthman Rasaq, **J. Waltermán**, *Stabilization of a Wave Using Non-collocated Observers and Boundary Feedback Controllers*.
2. A.K. Aydın, A.Ö. Özer, Uthman Rasaq, **J. Waltermán**, *Stabilization of a Wave Using Non-collocated Observers and Boundary Feedback Controllers*.

Presentations

1. *Diving into Wolfram Demonstrations Projects to Visualize the Controlled Partial Differential Equations*, AMS Southeastern Sectional, Poster Presenter, Savannah, GA, October 6th, 2024.
2. *Wolfram Demonstrations to Simulate Boundary Stabilization/Control of Certain Linear PDEs*, IEEE American Control Conference, Poster Presenter, Toronto, Canada, January 6th, 2024. <https://www.wku.edu/news/articles/index.php?view=article&articleid=11018>
3. *Wolfram Demonstrations to Simulate Boundary Stabilization/Control of Certain Linear PDEs*, AMS Joint Mathematics Meeting, Poster Presenter, Boston, MA, January 6th, 2023. <https://www.wku.edu/news/articles/index.php?view=article&articleid=11018>
4. *Wolfram Demonstrations Projects for the Feedback Sensor Design for a Multi-layer Sandwich Beam*, 42nd Annual WKU Mathematics Symposium, Bowling Green, KY, November 19th, 2022.
5. *Wolfram Demonstrations Simulating Boundary Stabilization/Control of Linear PDE's*, 53rd Annual WKU Student Scholar Showcase, Poster Presenter, Bowling Green, KY, April 1st, 2023.

6. *Uniformly Exponentially Stable Finite-Difference Model Reduction of a Heat and Piezoelectric Beam Interaction with Static or Hybrid Feedback Controllers*, 54th Annual WKU Student Scholar Showcase, Poster Co-Presenter, Bowling Green, KY, April 6th, 2024.
7. *Uniform Exponential Stability in Finite-Difference Model Reduction for Magnetizable Piezoelectric Beams with Non-Collocated Observers*, 54th Annual WKU Student Scholar Showcase, Poster Co-Presenter, Bowling Green, KY, April 6th, 2024.

Research Experience

- **Undergraduate Research Assistant, Western Kentucky University,** May 2022 – Present
 - Conducted control theory research on waves, beams, and piezoelectric spans under the supervision of Professor Ozer.
 - Worked on modeling and simulating the behavior of wave and beam systems with feedback control.
 - Explored control strategies to optimize the performance and stability of piezoelectric beams in various applications.
 - Developed simulations and analyzed results to support the development of theoretical models.

Professional Experience

- **Software Development Engineer Intern, Amazon, Arlington, VA** May 2023 – August 2023
 - Parsed and processed low-level system logs from prototype Amazon products.
 - Organized millions of time series data sets for detailed analysis and reporting.
 - Visualized data using Python libraries such as Matplotlib and Pandas to provide insights into product performance.
 - Collaborated with the engineering team to streamline data collection processes and optimize visualization pipelines.

Long-Term Substitute Teacher — Mathematics, Webster County Middle School (Clay, KY) — December 2024 – February 2025

Served as the sole mathematics instructor for a full class load during an extended teacher absence, functioning as teacher of record across multiple sections including 8th grade mathematics. Planned and delivered daily instruction, managed grades, and maintained classroom continuity independently for approximately two months. Gained direct experience teaching 8th grade math content to Kentucky public school students.

Substitute Teacher, Webster County Public Schools (Clay, KY) — February 2025 – Present

Continued as a substitute instructor at Webster County Middle School and Webster County High School following the long-term assignment, covering mathematics and other subject areas. Managed classrooms independently across middle and secondary grade levels, maintaining instructional expectations and behavioral standards.

Leadership & Extracurricular Activities

- **President, SIAM WKU Student Chapter** August 2022 – May 2023
 - Led the chapter in organizing events, including talks and seminars, aimed at promoting applied mathematics.
 - Organized a series of three talks by Wolfram engineers, providing students with insights into computational tools and their applications.
 - Managed chapter operations, including budgeting, event planning, and coordinating with faculty and students.
 - Facilitated collaborations between students and professionals in applied mathematics, enhancing the academic environment.
- **Wolfram Ambassador for Wolfram Mathematica** August 2022 – Present
 - Taught Mathematica coding to master's and undergraduate students, as well as faculty, focusing on computational methods and their applications in mathematics.
 - Delivered several talks at Western Kentucky University, demonstrating the use of Mathematica in solving complex mathematical problems.
 - Promoted the use of Mathematica as a key computational tool for both academic research and classroom learning.
 - Supported students and faculty in integrating Mathematica into their research and projects, expanding its usage across various fields.

Skills

- **Languages:** Python, Mathematica, Java, Kotlin
- **Libraries:** Pandas, Matplotlib
- **Developer Tools:** Git, Visual Studio Code, Vim, CLI, Linux

Membership

- American Mathematical Society.
- Society of Industrial and Applied Mathematics.